

SAMUEL MCKENZIE-SELL

SENIOR DATA ANALYST – GAME ENGINES & GAME DEV

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PROFESSIONAL SUMMARY

I am a versatile data analyst and game developer with a passion for transforming complex data into actionable insights and creating immersive digital experiences.

In the energy sector, I have led large-scale statistical analysis for Malaysia's largest electricity distributor, Tenaga Nasional Berhad (TNB), to assess the introduction of Time-of-Use tariff systems. I have built automated systems for analyzing and comparing energy company reporting in Australia using Python and SQL and constructed automated templates for companies to use for reporting for the AER.

Currently pursuing a game development degree at the University of Technology Sydney (UTS), I specialize in Unreal Engine, Unity, and custom C++/OpenGL engines. My projects include procedural generation for landscapes and cities, complex physics simulations, and computational geometry.

WORK EXPERIENCE

2017 – Present

Energy Market Consulting Associates

Senior Data Analyst (2021 – Present)

Lead a team to create standardised model templates

<https://www.aer.gov.au/industry/registers/resources/models/standardised-models-regulated-metering-services>

- Created standardised model templates for Capex/Opex/Pricing in *Excel* to provide greater transparency and clarity in regulatory reporting from energy companies along with documentation for the same.
- Lead project milestone sessions to report on progress to the regulating authority.
- Lead stakeholder feedback sessions to resolve conflicting requirements delivering a comprehensive and mutually beneficial reporting solution that met regulatory expectations.

Data Analyst (2017 – 2021)

Statistical analysis, Regulatory benchmarking

- Worked as the interface between the regulator and energy companies. Provided insights, justification, audits and analysis of company projects and expenditure to inform regulatory decisions on tariffs and pricing. Prioritized societal benefit, compliance with regulations and accuracy.

EDUCATION

University of Technology Sydney - Bachelor of Game Development (ongoing)

2022 - Communication for IT Professionals

2023 - Project Management and the Professional

2023 - Business Requirements Modelling

SKILLS

Programming Languages	C++, C#, GLSL, Javascript & WebGL, Python, Java, VBA
Engines	Unity, Unreal Engine
IDEs	VS, VSCode, Rider, CLion
Build & Tooling	CMake, Vite, Webpack, git
Data Handling	Web Scraping (Python), SQL, Data Flows (Tableau)
Data Analysis & Visualization	Python (Web Scraping, Data Modeling), SQL, Tableau, Office
Simulation	Procedural Generation, Position-Based Dynamics (PBDS), Custom Shaders (GLSL), Realtime Computational Geometry and Meshing